



Gonçalo Pascoal

Software Developer
Vestas

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- Porto, Portugal

About

Software engineer with experience in full-stack development, database management, cloud infrastructure and CI/CD, passionate about tackling difficult problems.

I hold a Master's degree in Informatics and Computing Engineering conferred by FEUP. My Master's dissertation leveraged deep reinforcement learning to compile quantum algorithms more efficiently for realistic architectures. I was distinguished with several awards for merit during my academic journey.

My main areas of interest include algorithms and data structures, optimization, low-level / systems programming, distributed systems, and performance-critical software. I consider myself to be rigorous, organized, and detail-oriented. I am also a hobbyist game developer and keenly interested in game design.

Languages

Portuguese Native
English Professional proficiency (C1/C2)
French Elementary proficiency (A1)

Hobbies

- Music (Guitar, Mandolin)
- Game Development
- Drawing

Experience

- Feb. 2025 – Present **Software Developer** Vestas
C# • Azure • ASP.NET • PostgreSQL • EF Core • Blazor • Python • Git REST APIs • Scrum
Simulation Development – Tower Structural Design Tool
- Complete architectural redesign of the Tower Structural Design Tool aimed at improving modularity, maintainability and traceability, as well as modernizing the technology stack.
 - Full rewrite of core business logic with greatly improved test coverage and reusability.
 - Implemented a flexible system for auditing structural calculations, allowing engineers to review the inputs, outputs and intermediate results of key procedures.
 - Completely overhauled data model, aligning it with business logic and domain knowledge to simplify schemas and reduce data duplication. Migrated from SQL Server to PostgreSQL, improving database performance and integrity using foreign keys, constraints, indices and denormalization.
 - Replaced legacy AngularJS frontend with interactive server-side Blazor, greatly improving UX, unifying tech stack under C# and streamlining deployment.
- Feb. 2024 – Feb. 2025 **Software Developer Trainee** Vestas
Python • Django • Azure • C# • SQL Server • Angular • Java • Git REST APIs • E2E Testing • Scrum
Simulation Development – Tower Structural Design Tool
- Full-stack development of new features, improvements, and bug fixes for a complex web application used for structural analysis, modeling, and design of wind turbine towers.
 - Contributed to the development and maintenance of CI/CD pipelines featuring build, near-zero downtime cloud deployment, testing, static analysis, and automatic versioning tasks.

Education

Faculty of Engineering, University of Porto (FEUP)
Porto, Portugal

- Sep. 2021 – Oct. 2023 **Master's Degree, Informatics and Computing Engineering**
Final Grade: 19.23 / 20
Thesis: *Noise-Adaptive Reinforcement Learning Strategies for Qubit Routing* (graded 20 / 20)
- Sep. 2018 – Jul. 2021 **Bachelor's Degree, Informatics and Computing Engineering**
Final Grade: 19.03 / 20

Awards / Grants / Scholarships

- 2024 **Prof. Doutor Raul Vidal / Deloitte Award** Deloitte
Granted to a FEUP M.EIC or M.ESW graduate that has distinguished themselves for the quality and innovation of their work in Software Engineering, and for their social, solidarity or student support activities
- 2023 **STSM Grant** COST (European Cooperation in Science and Technology)
Granted under [COST Action CA191935 – CERCIRAS](#) to visit the [SIMULA](#) research laboratory (Oslo, Norway) in the context of my M.Sc. thesis and discuss our methodology with other quantum computing researchers
- 2022 **Bondalti / Fundação Amália de Melo Award** Bondalti
For concluding the Bachelor's in Informatics and Computing Engineering at FEUP with the highest final grade
- 2021 **Merit Scholarship** DGES
For the average grade obtained during the 2019/2020 academic year
- 2020 **Merit Scholarship** DGES
For the average grade obtained during the 2018/2019 academic year
- 2020 **Prémio Incentivo / Incentive Award** University of Porto
For concluding the first year of the Bachelor's in Informatics and Computing Engineering at FEUP with the highest grade

Skills

Programming Languages

- **Most Experience:** Python, C#, SQL, Java
- **Experience:** C, C++, Rust, HTML, CSS, JavaScript, TypeScript, Dart
- **Some Experience:** PHP, Bash, Prolog

Technologies

Microsoft Azure, Git, Linux, Angular, React, LaTeX, Flutter, PyTorch, Qiskit, Godot Engine

Knowledge Areas

Deep Reinforcement Learning, Algorithms and Data Structures, REST APIs, Full-Stack Development

Other

Problem Solving, Autonomy, Resourcefulness, Time Management, Project Management, Leadership, Requirements Analysis, Technical Writing (English)

Publications

- Jul. 2024 **Deep Reinforcement Learning Strategies for Noise-Adaptive Qubit Routing** 
Gonçalo Pascoal, João Paulo Fernandes, Rui Abreu
2024 IEEE International Conference on Quantum Software ([IEEE QSW 2024](#))

Projects

- Master's Thesis**  Oct. 2023
Python • **PyTorch** • **Qiskit** • **Ray RLlib** • **NumPy** • **Pandas** • **LaTeX**
Deep Reinforcement Learning • **Quantum Compiling**
 - Leveraged deep RL (PPO) to compile quantum algorithms more efficiently for realistic architectures, helping to mitigate the adverse effects of noise on the outcome of computations.
 - Tackled the NP-complete qubit routing problem, which consists of inserting auxiliary operations to ensure that programs adhere to the connectivity constraints between qubits in a specific quantum architecture.
- Interactive Satellite Megaconstellation Simulation**  Jan. 2023
Rust • **Python** • **Godot Engine** • **Modeling and Simulation**
 - Analyzed effectiveness of different satellite connection strategies and orbital configurations for maintaining connectivity in the event of failures.
- Solver for Capacitated Vehicle Routing Problem**  Jul. 2022
C++ • **Data Structures** • **Map Matching** • **Search Algorithms** • **Metaheuristics**
 - Algorithms for solving large-scale CVRP instances (finding routes for a fleet of vehicles with multiple deliveries and limited carrying capacity). Implemented variants of popular metaheuristics found in the literature for CVRP (simulated annealing, ant colony optimization, granular tabu search).
 - Uses real-world OpenStreetMap data from Brazilian cities and performs map matching of GPS coordinates from test instances to graph vertices (using quadtrees or k-d trees).
- Unified Search System for Steam Games**  Jan. 2022
Apache Solr • **Python** • **Pandas** • **Data Processing & Analysis** • **Information Retrieval**
 - Aggregates Steam game data from multiple sources (public datasets, APIs, website scraping).
- Peer-to-Peer Distributed Backup Service**  Jun. 2021
Java • **Distributed Systems** • **Threads & Non-Blocking I/O** • **TCP Sockets w/ SSL**
 - Implements the Chord distributed hash table protocol. Files are divided into chunks stored across multiple peers.
 - Tackled scalability and fault tolerance concerns (thread pools, periodic tasks to manage peer failures).

Extra-Curricular Groups

- Oct. 2019 – Oct. 2023 **Tuna de Engenharia da Universidade do Porto**
Traditional academic group with over 35 years of history, bound by the values of music and friendship. Participating in the organization of events such as our festival (*PortusCalle*) has helped me develop and strengthen a diverse set of skills, such as multimedia, communication, teamwork, leadership, and working under time / resource pressure.